func _ready()

When instantiating a scene connected to the first executed scene, Godot will instantiate nodes down the tree (making _init() calls) and build the tree going downwards from the root.

This causes _enter_tree() calls to cascade down the tree. Once the tree is complete, leaf nodes call _ready. A node will call this method once all child nodes have finished calling theirs. This then causes a reverse cascade going up back to the tree's root.

_ready()

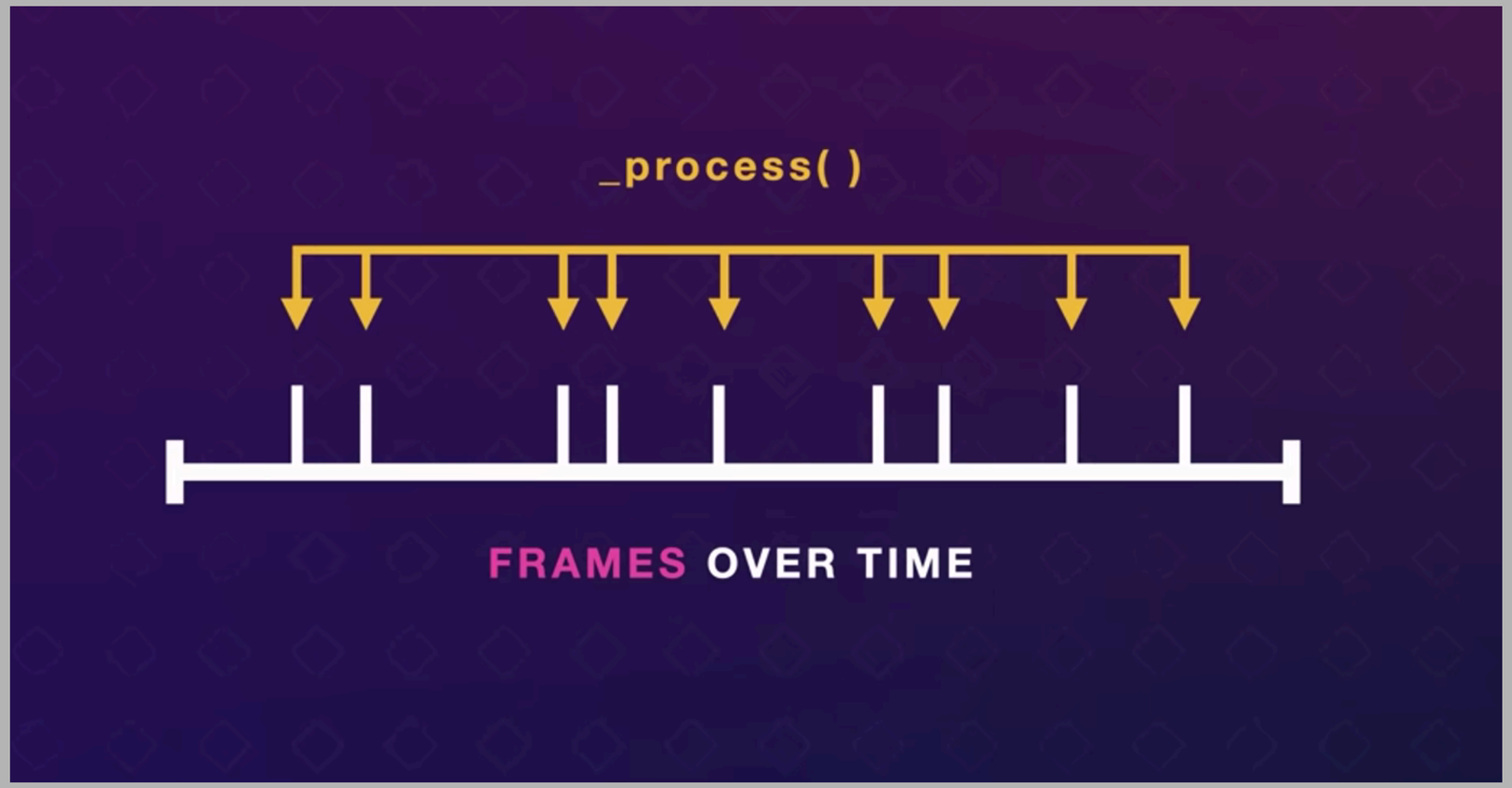
Games work by rendering many images per second, each called a frame, and they do so in a loop. We measure the rate at which a game produces images in Frames Per Second (FPS).

Most games aim for 60 FPS, although you might find figures like 30 FPS on slower mobile devices or 90 to 240 for virtual reality games.
The engine and game developers do their best to update the game world and render images at a constant time interval, but there are always small variations in frame render times. That's why the engine provides us with this delta time value, making our motion independent of our framerate.

func _process()

Use _process() when one needs a framerate-dependent delta time between frames.
If code that updates object data needs to update as often as possible, this is the right place.

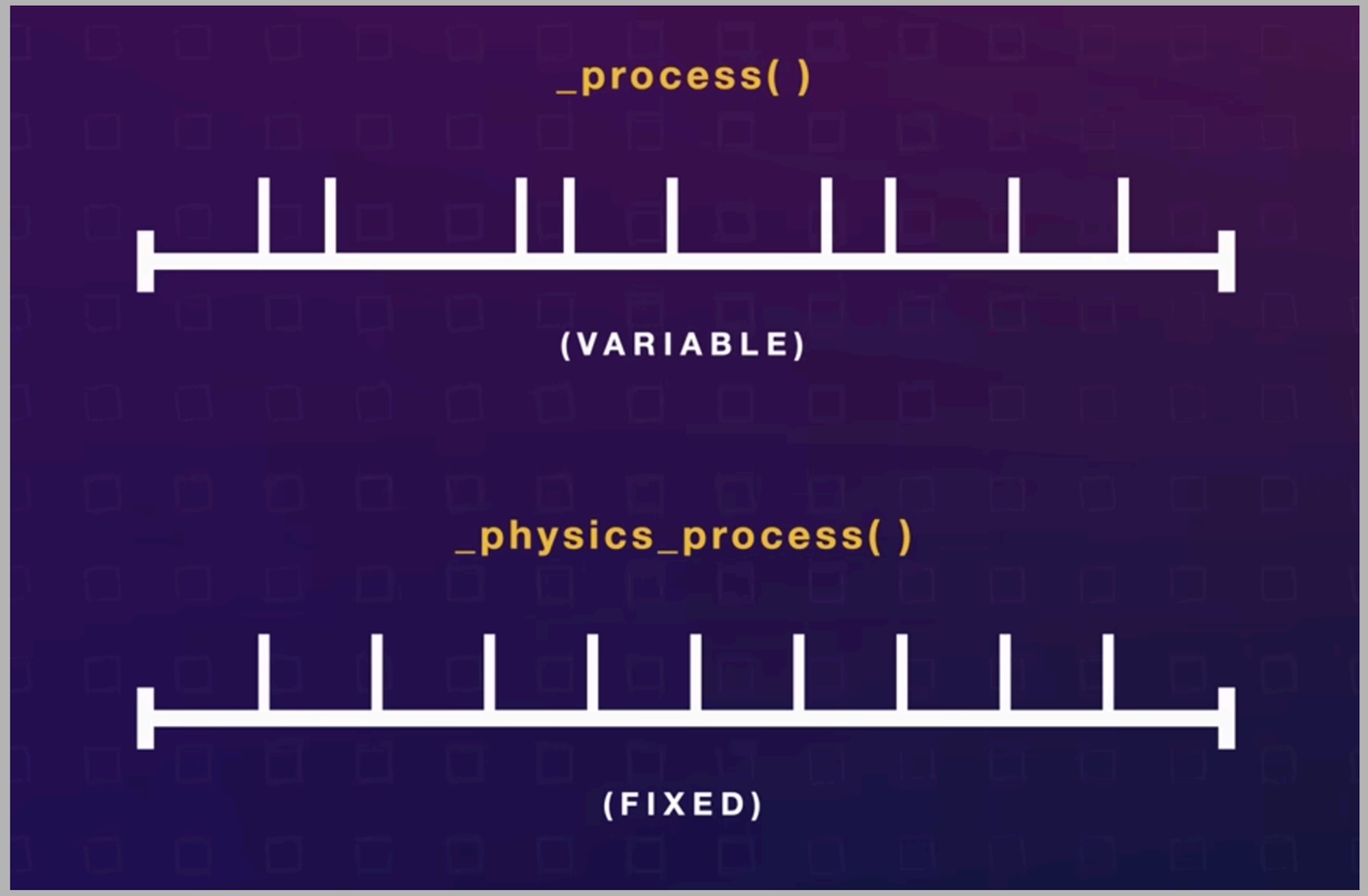
_process()



_physics_process()

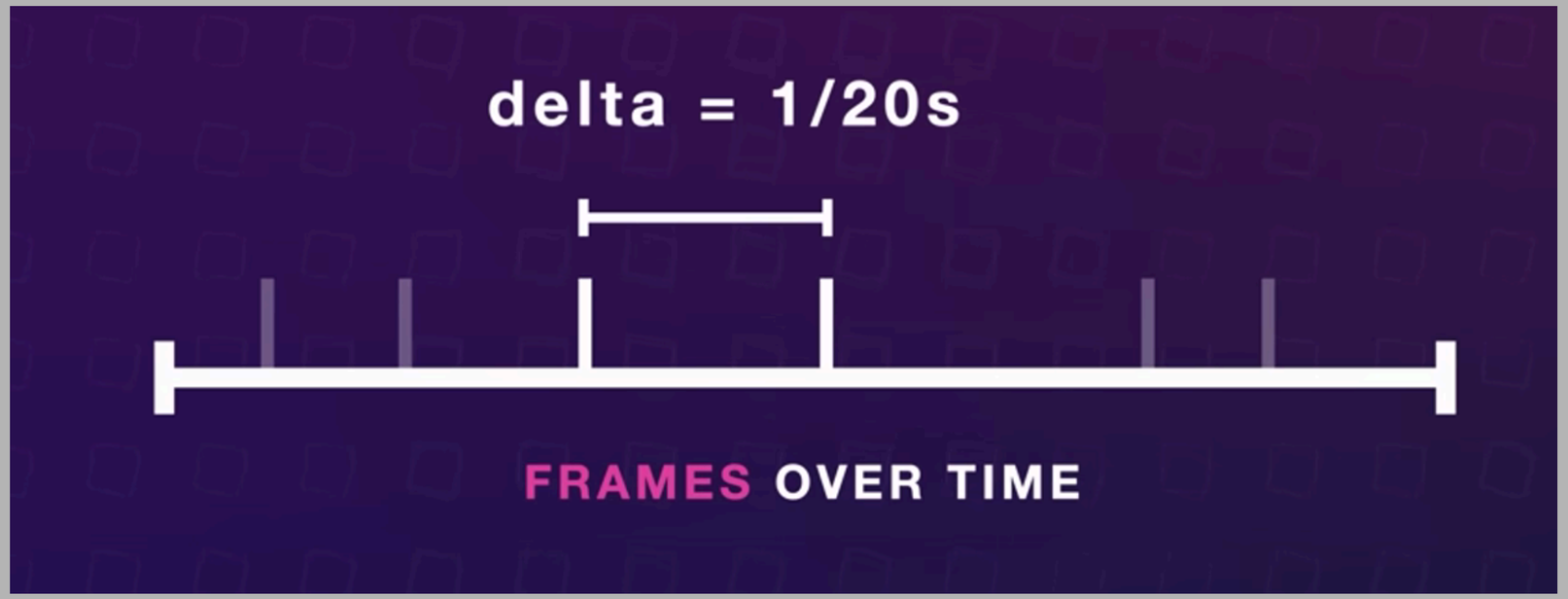
Use _physics_process() when one needs a framerate-independent delta time between frames.
If code needs consistent updates over time, regardless of how fast or slow time advances, this is the right place. Recurring kinematic and object transform operations should execute here.

_physics_process()



Delta

the time elapsed since the last frame

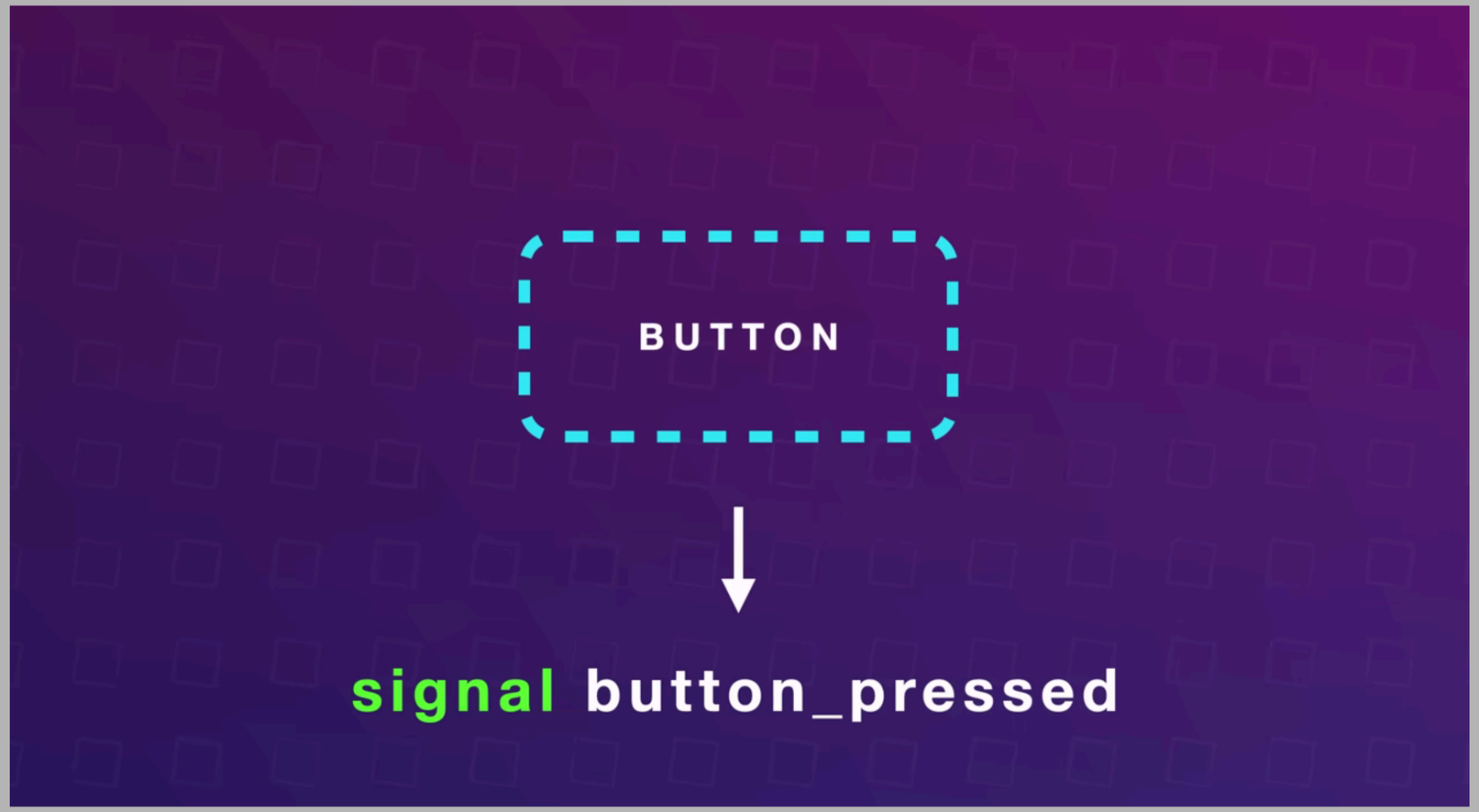


Signals

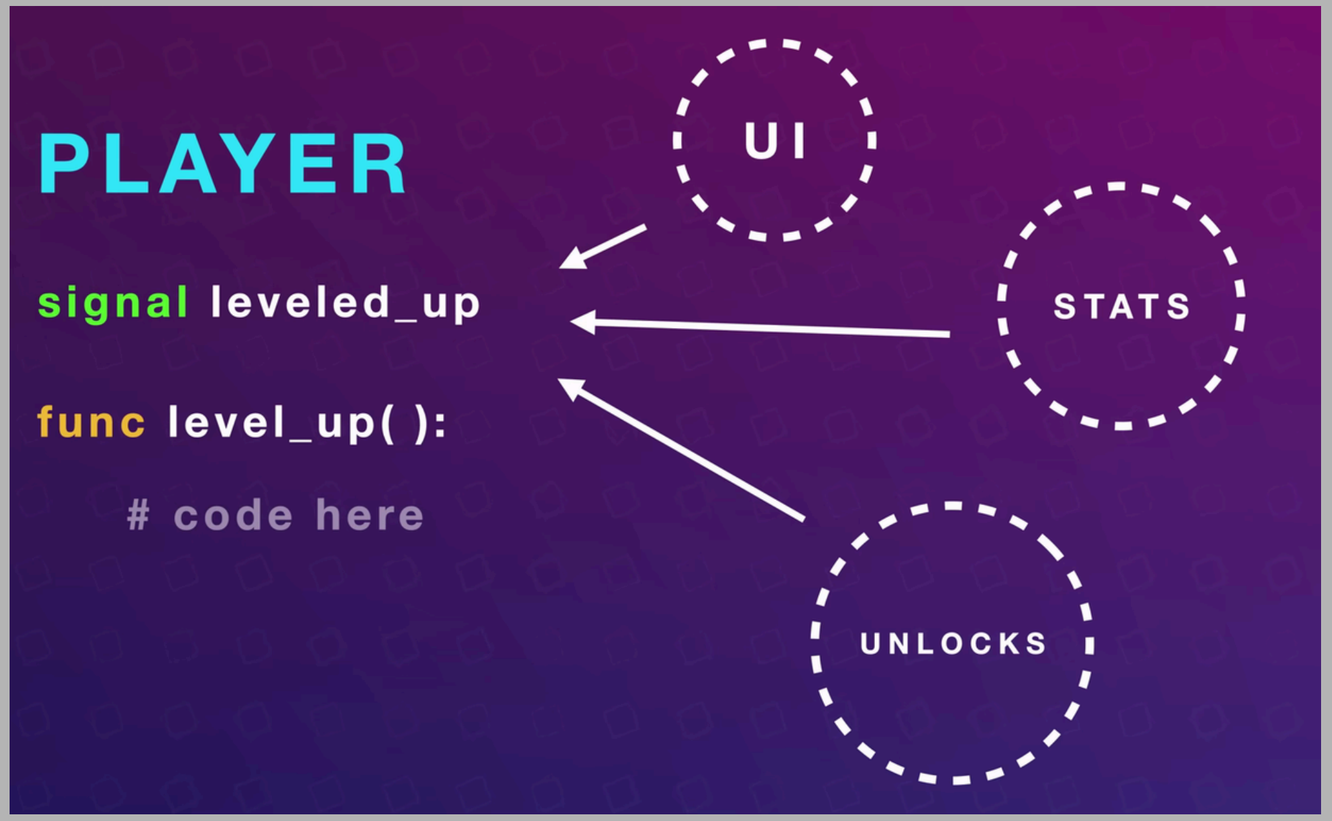
Signals are messages that nodes emit when something specific happens to them, like a button being pressed. Other nodes can connect to that signal and call a function when the event occurs.

Signals are a delegation mechanism built into Godot that allows one game object to react to a change in another without them referencing one another. Using signals limits coupling and keeps your code flexible.

Signal



Signals


PLAYER
signal leveled_up
func level_up( ):
code here
UI
STATS
UNLOCKS

Coyote time